Adaptive Colonoscopy Screening From a Mixed-Observability Markov Decision Process Built on the CMOST Microsimulation: More Cancers and Deaths Prevented per Colonoscopy Than Any Fixed Schedule

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Code

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Problem

> *Can adapting each person's next interval to their own findings beat the best fixed schedule at equal or lower colonoscopy volume?*

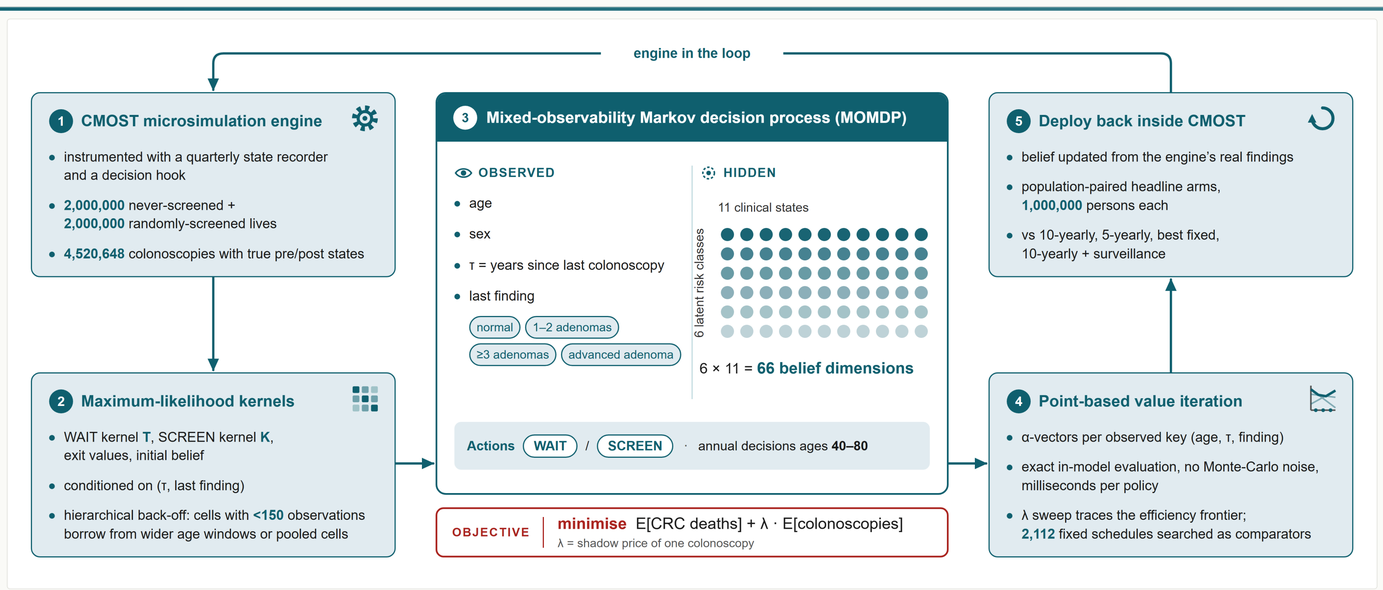
Colonoscopy prevents colorectal cancer (CRC) but is **scarce and invasive**: **judge it by what it prevents per exam.** Optimised schedules (Zaika et al. 2024) are **fixed** and ignore what a person's **colonoscopies found**, the one signal screening itself generates.

Motivation

- **A colonoscopy is evidence.** ≥3 adenomas lift the probability of the high-risk classes from **0.05** to **0.93** in men (0.98 in women); a normal exam lowers it to 0.007 in men and 0.017 in women.

23× higher lifetime CRC incidence in the top 2% of latent risk than in the bottom half: a gradient no fixed schedule can express, but one the policy infers from findings alone.

METHOD



CMOST supplies the kernels; point-based value iteration solves the MOMDP; policies run back inside the engine.

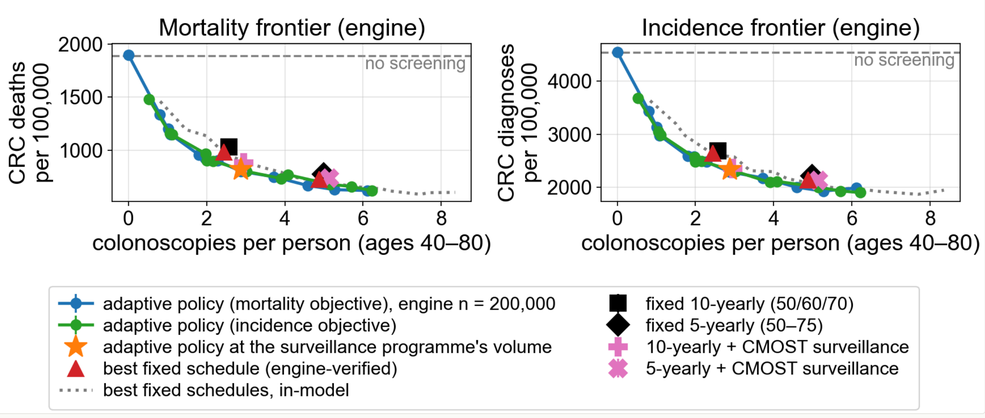
1 State. Observed: age, sex, years since last colonoscopy (τ), last finding (normal / 1–2 / ≥3 / advanced adenomas). Hidden belief: **6** risk classes × **11** clinical states. WAIT or SCREEN yearly, ages 40–80.

2 Kernels from the engine. Maximum-likelihood estimates from 4,000,000 CMOST lives (**4,520,648** colonoscopies), conditioned on (τ, last finding).

3 Solve, then deploy. Point-based value iteration; the shadow price λ sweeps the frontier. Every policy is re-run inside CMOST on its simulated findings against 10-yearly and 5-yearly screening, the best of 2,112 fixed schedules and 10-yearly + surveillance (headline arms: **1,000,000** persons).

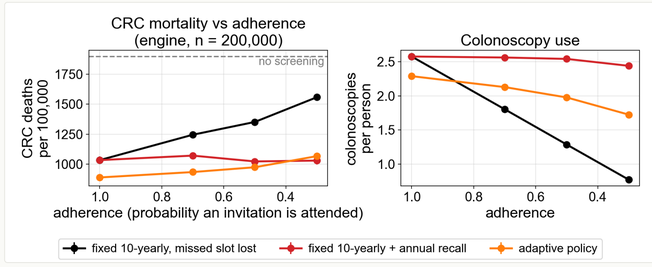
Key Results

PROGRAMME (ENGINE, 1,000,000 PER ARM)	COLONOSCOPIES PER PERSON	CRC DEATHS PER 100,000	DEATHS AVERTED VS NO SCREENING PER 1,000 COLONOSCOPIES
Fixed 10-yearly (50/60/70)	2.58	1032.1	3.32
Best fixed, no surveillance	2.44	980.9	3.71
Adaptive policy (ours)	**2.29**	**899.8**	**4.32**
10-yearly + surveillance	2.95	879.8	3.42
Adaptive, matched volume	2.87	818.1	3.72



Efficiency frontiers in the engine: CRC deaths (left) and diagnoses (right).

Robustness



Mortality and colonoscopy use vs adherence; n = 200,000.

18% mortality cut by fixed 10-yearly screening (missed slot lost) at 30% attendance, down from 46%; the adaptive policy, not re-solved, holds 44–53%.

Model ablation. Policies from three simpler models still beat 10-yearly screening by 121–172 deaths per 100,000 (n = 200,000).

Key numbers

+30%

CRC deaths averted per 1,000 colonoscopies vs 10-yearly screening

−132 CRC deaths per 100,000 vs 10-yearly

−201 CRC diagnoses per 100,000 vs 10-yearly

−11% colonoscopies per person vs 10-yearly

44–53% mortality cut at attendance 0.3–1.0 (n = 200,000)

−81 CRC deaths per 100,000 vs best fixed, no surveillance

+18% deaths averted per colonoscopy vs 5-yearly screening

What the policy does

COLONOSCOPY AGES ALONG EACH OBSERVATION PATH (λ = 0.001561)	MEN	WOMEN
all normal	49, 69, 78	58, 73
1–2 adenomas, then normal	49, 58, 67, 75	58, 63, 73, 80
≥3 adenomas, then normal	49, 50, 57, 64, 75	58, 59, 63, 69, 74, 79
advanced adenoma, then normal	49, 52, 64, 75	58, 63, 73, 80
advanced adenoma twice	49, 52, 53, 61, 67, 75	58, 63, 67, 71, 76, 80

Graded, immediate response: 1–2 adenomas bring the next exam in 9 years (men) or 5 (women); an advanced adenoma in 3 or 5.

SO WHAT → *A clean first exam earns 15–20 years; ≥3 adenomas pull the next exam to within 1–4 years (1 here); the sexes are offset by ~9 years (49 vs 58).*

Risk information vs adaptivity

If the risk class were **observable at baseline**, the policy would target it explicitly: **6.34 deaths averted per 1,000 colonoscopies,** none for the lowest-risk half and **7.49** for the highest-risk 5% (mortality 5,535 under 10-yearly to 1,956 per 100,000).

CRC DEATHS / 100,000 (IN-MODEL)	NO SCORE	SCORE (AUC 0.60)	SCORE BUYS
Best fixed schedule	972	942	−30
Adaptive policy	**892**	**867**	**−24**
Adaptivity buys	−80	−74	

3× more benefit from adaptivity than from a realistic baseline risk score; the two gains are nearly additive.

In the engine (n = 200,000) the two cannot be told apart (+6.5 ± 25.1 per 100,000).

Takeaway

Individualising colonoscopy timing on realised findings prevents more cancers and deaths per colonoscopy than any fixed schedule, including schedules optimised for the same simulator, and the gain survives imperfect adherence.

The solved policy is a legible three-tier surveillance rule with a sex offset: a 15–20-year interval after a clean first exam, 3–9 years after 1–2 or advanced adenomas, and re-examination within 1–4 years after ≥3 adenomas. Limitations: all results are internal to CMOST and inherit its calibration, and non-attendance is modelled as random. Life-years did not differ significantly between adaptive and fixed schedules, an honest null reported alongside the positive findings.